

Ordered Fields

D1.1 Field A field is a set F with addition and multiplication.

Addition Axioms Closure of Addition, Commutativity, Associativity, Identity Element, Additive Inverse

Multiplication Axioms Closure of Multiplication, Commutativity, Associativity, Identity Element, Multiplicative Inverse, Distributive Law

P1.4

- $x + y = x + z \Rightarrow y = z$
- $x + y = x \Rightarrow y = 0$
- $x + y = 0 \Rightarrow y = -x$
- $-(-x) = x$
- $x \neq 0 \wedge xy = xz \Rightarrow y = z$
- $x \neq 0 \wedge xy = x \Rightarrow y = 1$
- $x \neq 0 \wedge xy = 1 \Rightarrow y = 1/x$
- $x \neq 0 \Rightarrow 1/(1/x) = x$
- $0x = 0$
- $x \neq 0 \wedge y \neq 0 \Rightarrow xy \neq 0$
- $(-x)y = -(xy) = x(-y)$
- $(-x)(-y) = xy$

D1.5 Total Order A relation satisfying trichotomy and transitivity property. An ordered set is a set with a total order.

D1.8 Ordered Field A field where $y < z \Rightarrow x + y < x + z$,

$x > 0, y > 0 \Rightarrow xy > 0$. All ordered fields contain (a homomorphism of) $\mathbb{Q}$.

P1.12 Ordered Field Statements

- $x > 0 \Rightarrow -x < 0$ and vice versa
- $x > 0 \wedge y < z \Rightarrow xy < xz$
- $x < 0 \wedge y < z \Rightarrow xy > xz$
- $xy > 0 \Rightarrow (x > 0 \wedge y > 0) \vee (x < 0 \wedge y < 0)$
- $x \neq 0 \Rightarrow x^2 > 0$
- $0 < x < y \Rightarrow 0 < 1/y < 1/x$

Least Upper Bound Property

D1.16 Upper Bound Suppose S is an ordered set and $E \subset S$. If

$\exists \beta \in S, \forall x \in E, x \leq \beta$, then E is bounded above and β is an upper bound of E . Lower bound is defined similarly with $\geq$. They might not be unique and could possibly not exist. They might not be in the set itself.

D1.19 Least Upper Bound Suppose $\alpha \in S$, α is an upper bound of E , and if $\gamma < \alpha$ then γ is not an upper bound of E . Then α is the least upper bound or supremum of E , and $\alpha = \sup E$. We can similarly define the greatest lower bound or infimum $\alpha = \inf E$. The LUB is also unique.

R1.20 Maximum α is a maximum of E if it is a LUB and $\alpha \in E$.

T1.26 Least Upper Bound Property An ordered set S where $\forall E \subseteq S, E \neq \emptyset$, and E is bounded above implies $\exists \sup E \in S$. $\mathbb{Q}$ does not have this property.

T1.29 Completeness of $\mathbb{R}$ $\mathbb{R}$ is the ordered field which has the LUB property and is unique up to isomorphism, and contains $\mathbb{Q}$ as a subfield.

T1.32 An ordered set with the LUB property also has the GLB property.

P1.34 Let $E \subset \mathbb{R}, a \in \mathbb{R}, T = \{a + x : x \in E\}$. If E is bounded above, then $\sup T = a + \sup E$.

- $0 \neq E \subseteq \mathbb{R}, E$ is bounded above, so $\sup E$ exists
- $\forall x \in E, x \leq \sup E$
- $\forall x \in E, a + x \leq a + \sup E$
- $a + \sup E$ must be an upper bound for T , let M be any upper bound of T
- $\forall x \in E, a + x \leq M$
- $\forall x \in E, x \leq M - a$
- $M - a$ is an upper bound for E

• By definition of supremum,

$$\sup E \leq M - a, a > 0, \exists n \in \mathbb{N}, na > b \in F$$

R1.35 To show equality, either show $A \leq B \wedge B \leq A$ or $A \leq B, A < B$ is a contradiction

Archimedean Property

D2.1 Archimedean Property Let F be an ordered field. F has the Archimedean Property if $\forall a, b \in F, a > 0, \exists n \in \mathbb{N}, na > b \in F$

D2.1 Density For an ordered field $F, \mathbb{Q}$ is dense in F wrt ordering if

$$\forall x, y \in F, x < y, \exists r \in \mathbb{Q}, x < r < y \in F$$

P2.4 F has the Archimedean Property iff $\mathbb{Q}$ is dense in F

$\mathbb{Q}, \mathbb{R}$: $\mathbb{Q}$ is Archimedean and is dense in itself, while $\mathbb{R}$ is Archimedean as well

C2.6 $\forall \varepsilon \in \mathbb{R}_{>0}, \exists n \in \mathbb{N}, 0 < \frac{1}{n} < \varepsilon$

T2.7 nth roots $\forall x \in \mathbb{R}_{>0}, n > 0 \in \mathbb{Z}, \exists! y, y^n = x$

C2.8 For positive real a, b and positive integer $n, (ab)^{1/n} = a^{1/n}b^{1/n}$

C2.9 If $a, b \in \mathbb{R}, a < b, \exists x \in \mathbb{R} \setminus \mathbb{Q}, a < x < b$

P2.10 Let $\alpha \in \mathbb{R}, E = \{x \in \mathbb{Q} : x < \alpha\} \subseteq \mathbb{Q}$. Then $\sup E = \alpha$

Sequences

P2.13 Properties of absolute value

- $|a| \geq 0, a \leq |a|, -a \leq |a|$
- $|a| = 0 \Leftrightarrow a = 0$
- $|-a| = |a|$
- $|ab| = |a| \cdot |b|$
- $|a|^2 = a^2$
- $c \geq 0 \Rightarrow |a| \leq c \Leftrightarrow -c \leq a \leq c$
- $-|a| \leq a \leq |a|$

P2.14 Triangle Inequality $\forall a, b \in F, |a + b| \leq |a| + |b|$

C2.15 $\forall a, b \in F,$

- $||a| - |b|| \leq |a - b|$
- $|a - b| \leq |a| + |b|$
- $||a| - |b|| \leq |a + b|$

D2.16 Sequences A sequence in a set X is a function $x : \mathbb{N} \rightarrow X$, denoted by $(x_n)_n$ or $\{x_n\}_n$.

D2.21 Convergence A sequence in F converges iff

$\exists x \in F, \forall \varepsilon \in F_{>0}, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, n \geq N \Rightarrow |x_n - x| < \varepsilon$. For every small distance ε , there is a cut-off N such that for all larger n , the absolute difference is smaller than ε . x is the limit of the sequence in F , and $\lim_{n \rightarrow \infty} x_n = x$.

P2.23 The limit of a sequence is unique.

D2.24 Bounds A sequence is eventually constant if

$\exists N \in \mathbb{N}, \forall n \geq N, x_n = x_N = x$. A sequence is bounded in F if the set of values is bounded in F by some B . We can define one sided bounds similarly.

P2.26 If a sequence is convergent in F , then it is bounded.

E2.29 Example Convergence Proof

$$\begin{aligned} \left| \frac{2n^2 + 1}{n^2 + 3n} - 2 \right| &= \left| \frac{2n^2 + 1 - 2(n^2 + 3n)}{n^2 + 3n} \right| \\ &= \left| \frac{1 - 6n}{n^2 + 3n} \right| \\ &\leq \frac{1 + 6n}{n^2 + 3n} \\ &\leq \frac{n + 6n}{n^2} = \frac{7n}{n^2} = \frac{7}{n} \end{aligned}$$

• Let $\varepsilon > 0$ be given, then choose $K \in \mathbb{N}$ such that $K > \frac{7}{\varepsilon}$. Then $\forall n \geq K,$

$$\begin{aligned} \left| \frac{2n^2 + 1}{n^2 + 3n} - 2 \right| &\leq \frac{7}{n} \text{ since } n \geq K \\ &\leq \frac{7}{K} \\ &< \varepsilon \text{ since } K > \frac{7}{\varepsilon} \end{aligned}$$

Convergent Sequences

P3.1 Properties of Limits of Convergent Sequences Let $(s_n)_n, (t_n)_n$ be convergent sequences in F . Then $(s_n + t_n)_n, (cs_n)_n, (s_nt_n)_n, (|s_n|)_n, (\sqrt{s_n})_n, (\frac{1}{s_n})_n$ are all convergent with limits following the substitution. If $\exists N \in \mathbb{N}, \forall n \geq N, s_n \leq t_n$, then $\lim_{n \rightarrow \infty} s_n \leq \lim_{n \rightarrow \infty} t_n$.

P3.2 Squeeze Theorem If $(x_n)_n, (y_n)_n, (z_n)_n$ are sequences in an ordered field F such that $\forall n, x_n \leq y_n \leq z_n$ and $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} z_n = a$, then $\lim_{n \rightarrow \infty} y_n = a$.

More Sequences

E3.3 For real $0 \leq b < 1, \lim_{n \rightarrow \infty} b^n = 0$.

E3.6 For real $c > 0, \lim_{n \rightarrow \infty} c^{\frac{1}{n}} = 1$.

E3.7 For real $a > 1$ and integer $k > 0, \lim_{n \rightarrow \infty} \frac{n^k}{a^n} = 0$.

D3.9 Monotone Sequences (s_n) is monotonically increasing if $\forall n, s_n \leq s_{n+1}$, while it is monotonically decreasing if $\forall n, s_n \geq s_{n+1}$.

T3.10 Monotone Convergence Every monotone sequence in $\mathbb{R}$ is bounded iff it converges.

Subsequences, Cauchy Sequences

D4.2 Subsequence Given a sequence in F , consider $(n_k)_k$ of positive integers where $n_1 < n_2 < \dots < n_k < \dots$. Then $(x_{n_k})_k$ is a subsequence of $(x_n)_n$.

E4.3 A sequence converges to x iff every subsequence also converges to x . For divergence, we can show a subsequence which diverges, or two subsequences which converge to different limits.

T4.9 Existence of Monotone Subsequence Let there be a sequence in an ordered field F which is not necessarily convergent. There exists a monotone subsequence in it.

T4.10 Bolzano-Weierstrass Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

D4.11 Cauchy Sequence A sequence $(p_n)_n$ in F is said to be Cauchy if

$$\forall \varepsilon \in F_{>0}, \exists N \in \mathbb{N}, \forall m, n \in \mathbb{N}_{\geq N}, |p_n - p_m| < \varepsilon.$$

P4.13 If a sequence is convergent, then it is Cauchy.

P4.15 If a sequence is Cauchy, then it is bounded.

P4.16 Properties of Cauchy Sequences If $(s_n), (t_n)$ are Cauchy sequences, then $(s_n + t_n), (cs_n), (s_nt_n), (|s_n|)$ are also Cauchy.

More Cauchy Sequences

D4.18 Cauchy Complete An ordered field in F is Cauchy complete iff every Cauchy sequence in F is convergent in F . $\mathbb{R}$ is Cauchy complete.

P4.19 Let p_n be a Cauchy sequence in F . If there exists a subsequence of p_n that converges in F , then p_n also converges in F to the same limit.

T4.20 $\mathbb{R}$ is Cauchy complete.

D4.22 Contractive A sequence x_n is contractive if there exists C with $0 < C < 1$ such that $|x_{n+2} - x_{n+1}| \leq C|x_{n+1} - x_n|$ where C does not depend on n .

P4.23 Every contractive sequence is Cauchy and thus convergent.

D4.24 Extended Real Numbers The extended real number system consists of $\mathbb{R}, +\infty$, and $-\infty$. The original order is preserved and $\forall x \in \mathbb{R}, -\infty < x < +\infty$. This ordered set (not field) is denoted by $[-\infty, +\infty]$. The supremum and infimum of any subset always exists.

D4.28 Limit to Infinity The limit of x_n is $+\infty$ if for all real $M, \exists N, \forall n \geq N, s_n \geq M$. Similarly for $-\infty$.

P4.30 Let x_n be a sequence in $\mathbb{R}, x \in [-\infty, +\infty]$. Then $x_n \rightarrow x$ iff $\forall A \in [-\infty, \infty], A < x, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, n \geq N \Rightarrow A < x_n$ and the opposite holds.

L4.31 For any sequence $x_n \in \mathbb{R}$, there exists a subsequence x_{n_i} which converges in $[-\infty, +\infty]$.

limsup, liminf

D5.1 limsup, liminf Let x_n be a sequence in $\mathbb{R}$. Let $E = \{\text{the limits in } [-\infty, +\infty] \text{ of all convergent subsequences of } x_n\}$. Then define $\limsup_{n \rightarrow \infty} = \sup(E), \liminf_{n \rightarrow \infty} = \inf(E)$. $\limsup$ is the limit superior, while $\liminf$ is the limit inferior.

P5.3 Equivalent Definition Let x_n be a bounded series in $\mathbb{R}$ and $M = \limsup x_n$. $M = \limsup x_n$ if $M \in \mathbb{R}$ satisfies the following: For each $\varepsilon > 0$ there are at most finitely many n such that $x_n \geq M + \varepsilon$. Alternatively, $\forall \varepsilon, \exists K \in \mathbb{N}, \forall n \geq K, x_n < M + \varepsilon$. Also, for each $\varepsilon > 0$, there are infinitely many n such that $x_n > M - \varepsilon$. The same holds for $\liminf$.

P5.5 Convergence with limsup, liminf Let s_n be a sequence in $\mathbb{R}$. Then $\liminf_{n \rightarrow \infty} s_n \leq \limsup_{n \rightarrow \infty} s_n \in [-\infty, +\infty]$. Equality only holds iff s_n converges.

P5.7 If $s_n \leq t_n, \forall n \geq N, \liminf_{n \rightarrow \infty} s_n \leq \liminf_{n \rightarrow \infty} t_n$ and $\limsup_{n \rightarrow \infty} s_n \leq \limsup_{n \rightarrow \infty} t_n$

P5.8 Equivalent Definition of limsup Let x_n be a bounded sequence. For each $n \in \mathbb{N}$, let $y_n = \sup\{x_k : k \geq n\}$. Then $\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n$.

P5.9 Let x_n be a bounded sequence. For each $n \in \mathbb{N}$, let $z_n = \inf\{x_k : k \geq n\}$. Then z_n is increasing and bounded and $\liminf x_n = \lim_{n \rightarrow \infty} z_n$

Series

D5.10 Series Let F be an ordered field. Given a sequence $a_n \in F$, we can form an infinite series $\sum_{k=1}^{\infty} a_k = a_1 + a_2 +$ More precisely, define $S_k = \sum_{n=1}^k a_n$ (nth partial sum).

P5.13 Properties of Convergent Series If $\sum a_n, \sum b_n$ are convergent series and $c \in F$, then $\sum(a_n + b_n)$, and $\sum ca_n$ are convergent.

P5.14 Cauchy Criterion for Series Let a_n be a sequence in $\mathbb{R}$. Then $\sum a_n$ converges iff $\forall \varepsilon > 0, \exists N, \forall m, n, m \geq n \geq N, |\sum_{k=n}^m a_k| \leq \varepsilon$.

C5.15 If $\sum a_n$ converges, $\lim_{n \rightarrow \infty} a_n = 0$. However, the converse is false.

C5.17 nth term test If a_n does not converge or converges to a nonzero, then $\sum a_n$ diverges.

D5.19 A series eventually has a property (*) if $\exists N \in \mathbb{N}, \forall k \geq N, a_k$ has property (*).

P5.20 Let $\sum a_n$ be an eventually non-negative series where $a_n \in \mathbb{R}$. Then a_n converges iff S_n is bounded above.

E5.22 If $p > 1$, the p-series $\sum \frac{1}{n^p}$ converges. When $p \leq 1$, the series diverges.

D5.23 Absolute Convergence The series $\sum a_n$ converges absolutely if $\sum |a_n|$ converges. If a series converges but not absolutely, it converges conditionally.

P5.24 Let a_n be a sequence in $\mathbb{R}$. If $\sum a_n$ converges absolutely, then a_n converges. However, the converse is false.

Convergence Tests

P6.1 Direct Comparison Test Let a_k, b_k be sequences in $\mathbb{R}$ such that $\sum a_k, \sum b_k$ are eventually non-negative. Suppose we eventually have that $0 \leq a_k \leq b_k$. If $\sum b_k$ converges, then $\sum a_k$ converges. If $\sum a_k$ diverges, then $\sum b_k$ diverges.

P6.4 Limit Comparison Test Let a_k, b_k be sequences in $\mathbb{R}$ such that $\sum a_n, \sum b_n$ are eventually positive, and suppose the limit $\rho = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ exists. If $\rho > 0$, they either both converge or both diverge. If $\rho = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.

P6.5 Ratio Test Let a_n be a sequence in $\mathbb{R}$ such that $\sum a_n$ is eventually positive, and let $\rho = \limsup \frac{a_{n+1}}{a_n}$. If $\rho < 1, \sum a_n$ converges. If $\exists N \in \mathbb{N}, \forall n \geq N, \frac{a_{n+1}}{a_n} \geq 1$, then the series diverges.

P6.8 Root Test Let a_k be a sequence in $\mathbb{R}$ such that $\sum a_n$ is eventually non-negative, and let $\rho = \limsup a_n^{1/n}$. If $\rho < 1$, then $\sum a_n$ converges. If $\rho > 1$, then $\sum a_n$ diverges. If $\rho = 1$, we cannot make a conclusion.

L6.9 For any sequence c_n of positive numbers, $\liminf_{n \rightarrow \infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \limsup_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \limsup_{n \rightarrow \infty} \frac{c_{n+1}}{c_n}$.

T6.11 Dirichlet's Test Let a_n, b_n be sequences in $\mathbb{R}$. Suppose the partial sums A_n form a bounded sequence, b_n is monotone decreasing, and $\lim b_n = 0$. Then $\sum a_n b_n$ converges.

L6.12 Partial Summation Given a_n, b_n , indexed by $Z \geq 0$, define $A_{-1} = 0$ and $\forall n \geq 0, A_n = \sum_{k=0}^n a_k$. Then for any $0 \leq p \leq q$, we have $\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q B_q - A_{p-1} b_p$.

C6.14 Alternating Series Test Let $\sum (-1)^{n+1} b_n$ or $\sum (-1)^n b_n$ be an alternating series with b_k as a sequence in $\mathbb{R}$. Suppose $b_n \geq 0, b_n$ is monotone decreasing, and $\lim_{n \rightarrow \infty} b_n = 0$. Then it is convergent.

Rearrangement of Series

D6.15 Rearrangement Let σ be a permutation of $\mathbb{N}$. Then the series $\sum_{n=1}^{\infty} a_{\sigma(n)}$ is a rearrangement of $\sum_{n=1}^{\infty} a_n$.

T6.16 Rearrangement of Absolutely Convergent Series Let a_n be a sequence in $\mathbb{R}$, and $\sum a_k$ be an absolutely convergent series. Then any rearrangement of $\sum a_k$ also converges and has the same sum as $\sum a_k$.

T6.17 Riemann Rearrangement Let a_k be a sequence in $\mathbb{R}$ such that $\sum a_n$ is conditionally convergent. Let $\alpha, \beta \in [-\infty, +\infty], \alpha \leq \beta$. Then there exists a rearrangement $\sum b_n$ of $\sum a_n$ such that the sequence of partial sums s'_n of b_n satisfies $\liminf s'_n = \alpha, \limsup s'_n = \beta$.

Random Tutorial Crap

Bernoulli's Inequality $\forall x \geq -1, \forall n \in \mathbb{N}, (1+x)^n \geq 1+nx$

T1P7 $\sup X = -\inf -X$

T2P4 $\lim_{n \rightarrow \infty} |x_n| = 0$ iff $\lim_{n \rightarrow \infty} x_n = 0$.

T3P5 x_n converges to $x \in \mathbb{R}$ iff x_{n_k} converges to x for every subsequence of x_n .

T3P6 If every subsequence of x_n has a subsequence which converges to 0, then x_n converges and $\lim_{n \rightarrow \infty} x_n = 0$.

T4P1 $\lim_n \cos n$ does not exist.

T4P6 $\limsup(x_n + y_n) \leq \limsup x_n + \limsup y_n$, but equality might not hold.